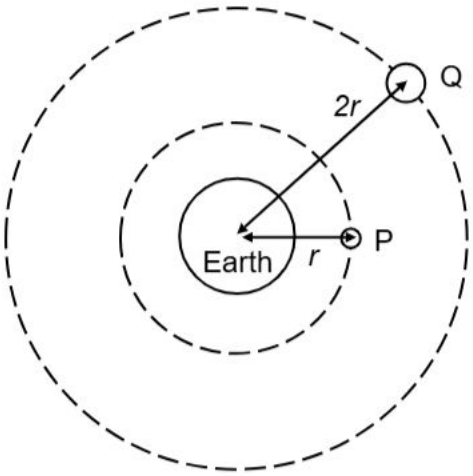
8 Two satellites P and Q are in circular orbits about the Earth. Satellite P has mass m and it is in an orbit of radius r. Satellite Q has mass $2m$ and it is in an orbit of radius $2r$.



Which of the following statements is true?

A The speed of satellite Q is greater than that of satellite P.

B The speed of satellite Q is the same as that of satellite P.

C The kinetic energy of satellite Q is greater than that of satellite P.

D The kinetic energy of satellite Q is same as that of satellite P.

Ans: D

In a stable orbit of radius d,

Gravitational force provides centripetal force

$$\frac{GMm}{d^2} = \frac{mv^2}{d}$$

Hence, ke = $\frac{1}{2}mv^2 = \frac{GMm}{2d}$

and $v = \sqrt{\frac{GM}{d}}$

$KE_P = \frac{GMm}{2r}$ and $KE_Q = \frac{GM(2m)}{2(2r)} = \frac{GMm}{2r}$

$\Rightarrow KE_P = KE_Q$

$v_P = \sqrt{\frac{GM}{r}}$ and $v_Q = \sqrt{\frac{GM}{2r}}$

$\Rightarrow v_P > v_Q$